VoiceOps 2026: Capstone Technical Report

Project 4: Policy-Aware Structured Decision Agent | Author: Person 4 | IIT Guwahati × Gnani.ai

1. Executive Summary & Architecture

PERSON 4 implements a **Policy-Aware Structured Decision Support Agent** designed around strict regulatory compliance, deterministic decision guardrails, and an explicit finite state machine. To eliminate unauthorized commitments and policy hallucinations, the architecture separates intent classification, policy retrieval, decision validation, and tool execution into distinct stages. The conversation flow is strictly governed by a **DialogStateMachine** enforcing identity verification retry limits (max 1 retry) and gating private order disclosures. Business day timelines are calculated deterministically against the frozen reference date **20 October 2026**, skipping official warehouse holidays. Claims exceeding ■10,000 for damaged items are routed to human Priority Resolution.

Component	Implementation Mechanism	Key Design Distinction
Intent Router	Structured intent extraction into CustomerIntent dataclass	Decouples query classification from retrieval and synthesis
State Machine	Finite State Machine (FSM) with 6 explicit states	Strict verification gating and human escalation locking
Decision Guardrails	Threshold checks (>■10,000 damage claim) & precedence gate	Prevents unauthorized AI approvals; mandates human review
Calendar Engine	Deterministic business-day math skipping Sundays and holidays	Accurate date calculations using reference date 20 Oct 2026
Voice Pipeline	Official Gnani STT and TTS SDK (Kaveri, timbre-v2.5)	Batch audio folder processing with execution audit logs

2. Implementation of the Four Official Steps

Step 1 — Baseline: Implemented in src/baseline.py. Standard prompt-only configuration identifying as an AI assistant. Scores 88.5% on broad decisions but achieves only 13.5% fact coverage due to ungrounded guessing.

Step 2 — Policy RAG: Implemented in src/rag_agent.py and src/retriever.py. Employs structured intent extraction to route directly to authoritative policy documents. Adheres to KB-00.1 precedence hierarchy (Policy > Guide > Help > FAQ > Marketing) and validates effective dates.

Step 3 — Conversational Agent and Tools: Implemented in src/conversation_agent.py and src/state_machine.py. Multi-turn dialogue manager enforces identity verification (max 1 retry), business day calendar math (20 October 2026 reference date, skipping Sundays and 19 October warehouse holiday), and ticket escalation for high-value claims (>■10,000).

Step 4 — Voice with Gnani.ai: Implemented in src/voice_adapter.py. Connects Gnani STT and TTS directly to the Step 3 agent. Batch processing executes reliably via python run.py --input-dir ... --output-dir ... with 100% test completion.

3. Empirical Evaluation Results

Evaluation Metric	Step 1: Baseline	Step 2: Policy RAG	Step 3: Multi-Turn	Compliance & Quality
Decision Accuracy	88.5%	88.5%	86.5%	State-constrained decisions
Fact Coverage Rate	13.5%	53.8%	54.2%	+40.3% factual accuracy
Gold Evidence Recall	0.0%	65.4%	91.2%	Section ID provenance verified
Tool-Call Accuracy	N/A	N/A	92.7%	Strict verification gates
FSM State Violations	N/A	0	0	100% legal transitions

4. Technical Analysis: What Worked & What Failed

Strengths of Structured Policy Decisions:

1. **Zero Unauthorized Approvals:** In high-value damage scenarios (e.g. ■20,000+ TVs), the FSM strictly intercepted replacement requests, enforced ticket logging, and barred the assistant from promising approvals.
2. **Audit Trail Transparency:** Every turn produced structured intent, entity dictionaries, and transition logs, allowing full compliance auditing.
3. **Calendar Accuracy:** Deterministic date math prevented hallucinations on holiday exclusions (e.g. 19 October festival closure).

Challenges & Mitigations:

1. **Multi-Intent Messages:** Customers expressing multiple issues in one sentence required hierarchical intent prioritization.
2. **Strict State Transitions:** Edge cases where customers jumped topics mid-verification were safely handled by offering general policy answers without revealing order records.

5. Limitations and Future Directions

• **Dynamic Policy Graph:** Migrating tabular precedence rules into an executable policy knowledge graph (e.g. RDF/OWL).
• **Automated Fraud Detection:** Enhancing the identity verification state machine with device fingerprinting and velocity checks.